Abdelrahman Elsawy

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Education

Technische Universität Dortmund Oct 2022 - 2027 (Expected)

Bachelor of Science (B.Sc.) in Computer Science *Dortmund, Germany*

- Relevant topics include algorithms, data structures, probability, linear algebra, statistics, and machine learning.
- Building personal projects in AI agents, retrieval-augmented generation, and persistent conversational memory.
- Expected to complete the B.Sc. in 2027 and planning to continue with a Master's degree.

Projects

Bachelor's Thesis Assistant 2026

AI Agents & Retrieval-Augmented Generation *Repository Link*

- Built a LangGraph workflow coordinating Planner, Worker, and Critic roles, with human approval gates and SQLite checkpoints for resumable research sessions.
- Implemented PDF text extraction, embedding-based retrieval, and reranking to support cited drafts; used LangChain integrations and Pydantic schemas for structured model responses.
- Connected a FastAPI backend to a web interface with streamed progress and report downloads; published a separate scripted browser demo on GitHub Pages.

Mem0 Customer Memory 2026

Persistent Conversational Memory *Repository Link*

- Built a Streamlit support chat that retrieves customer-specific memories before generating replies and preserves stored facts across new chat sessions.
- Integrated Mem0, local Qdrant, and FastEmbed for memory storage and search, Groq for replies and fact extraction, and SQLite for stable customer IDs.
- Added views of saved and retrieved memories, customer-filtered searches, and memory deletion to explore how persistent context affects replies.

Skills

Programming:	Python, SQL
AI Applications:	LangChain, LangGraph, RAG, Mem0, Qdrant, FastEmbed
Backend & Interfaces:	FastAPI, Pydantic, SQLite, Streamlit
Tools:	Git, GitHub, pytest, Jupyter Notebook, VS Code
ML Foundation:	Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch
Currently Learning:	MCP (Model Context Protocol), Docker
Languages:	German (B2), English (B2)